Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 30 epochs with a batch size of 128.

2 Results

Figure 1 shows the test accuracy of BERT-base by dataset.

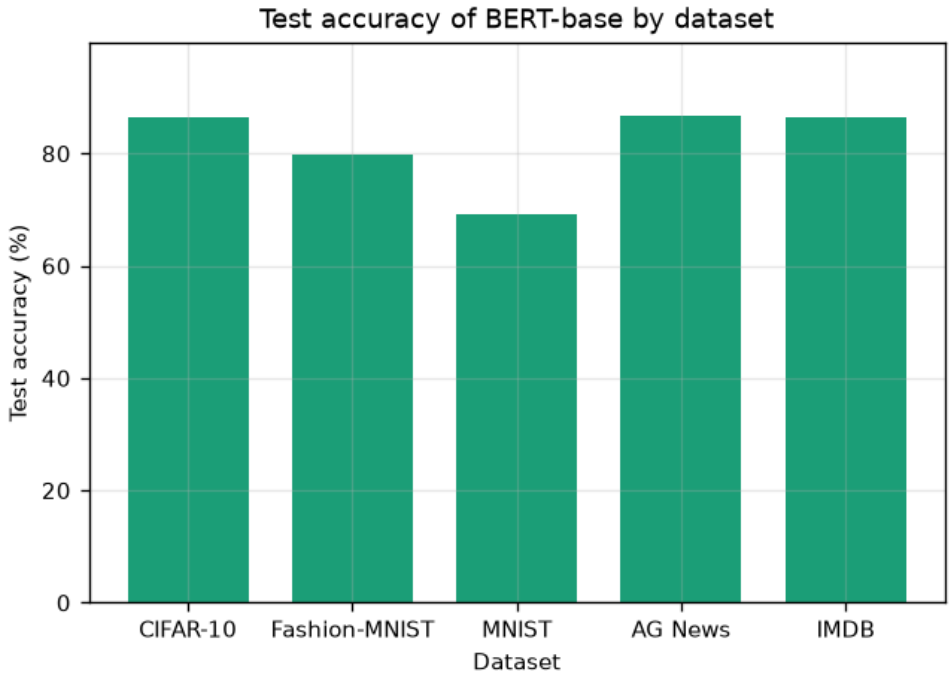


Figure 1: Test accuracy of BERT-base by dataset for the models in this experiment.

BERT-base reaches 86.7% test accuracy on IMDB. BERT-base achieves 66.7% test accuracy on IMDB (Figure 1).

Figure 2 shows the f1 score during training.

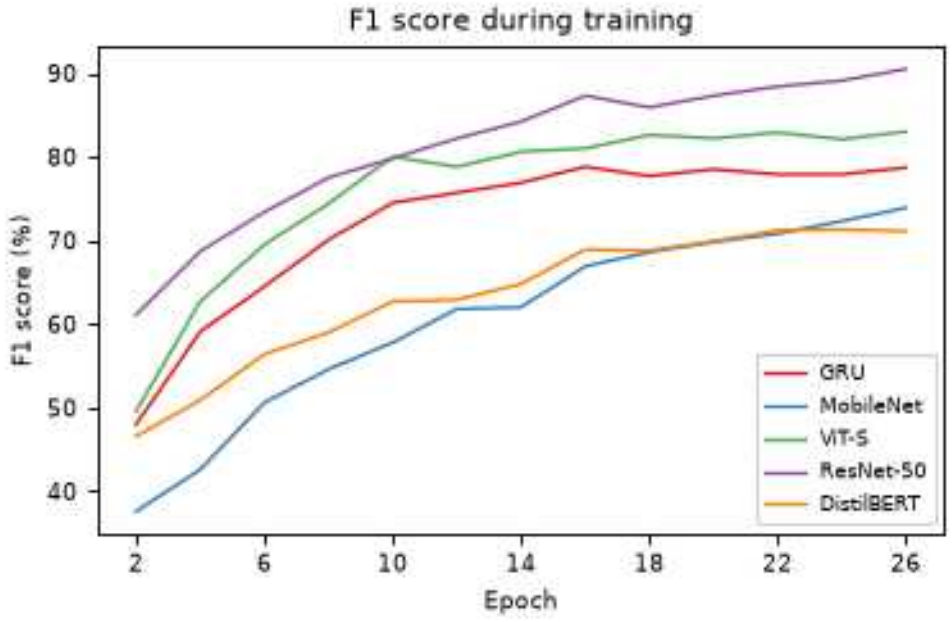


Figure 2: F1 score during training for the models in this experiment.

ResNet-50 attains a F1 score of 49.6% at epoch 2. ViT-S reaches 85.8% F1 score at epoch 22 (Figure 2). At epoch 22, MobileNet achieves 65.0% F1 score. GRU attains a F1 score of 81.7% at epoch 8.

Figure 3 shows the test accuracy of CNN by dataset.

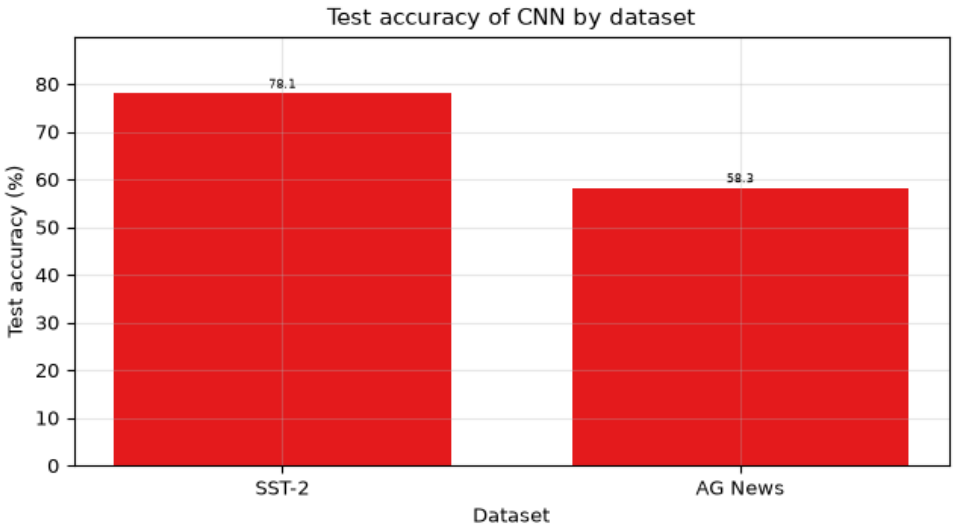


Figure 3: Test accuracy of CNN by dataset for the models in this experiment.

As shown in Figure 3, CNN scores 28.8 points higher on SST-2 than on AG News. CNN scores 1.8 points higher on SST-2 than on AG News. CNN scores 19.8 points higher on SST-2 than on AG News.

Figure 4 shows the validation accuracy during training.

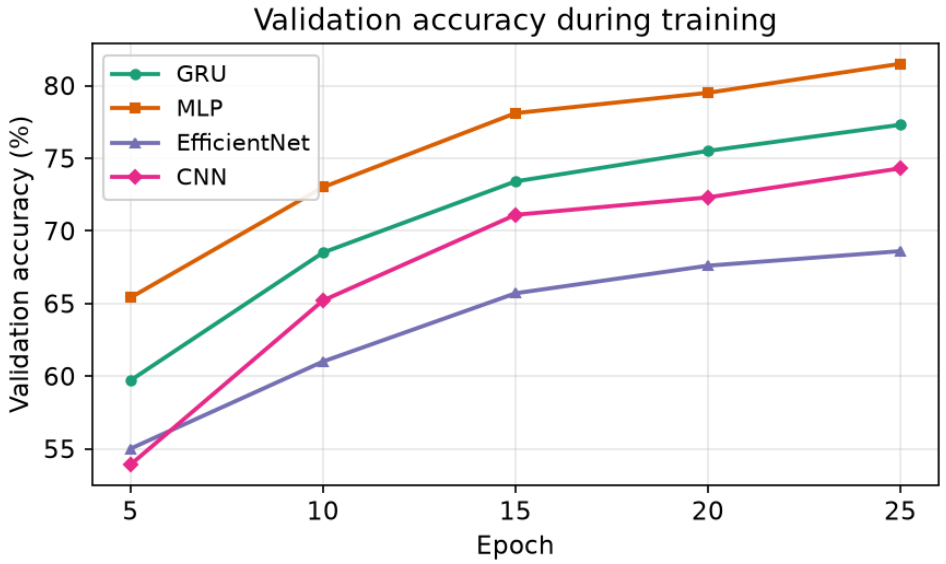


Figure 4: Validation accuracy during training for the models in this experiment.

From epoch 10 to epoch 20, the validation accuracy of CNN improves by a relative 10.9%. CNN outperforms EfficientNet by 6 percentage points at epoch 25 (Figure 4). CNN reaches 38.7% validation accuracy at epoch 5 (Figure 4). As shown in Figure 4, at epoch 10, EfficientNet achieves 61.0% validation accuracy.

3 Discussion

SVM reaches 88.8% accuracy on Yelp. We note that the training was stable for all models.